

# **Mid Way Report**

**on**

## **AiElectron**



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## **1. Introduction**

The continuous rise of electric vehicles (EVs), along with advancements in IoT and smart energy technologies, has created new opportunities in the utility and mobility sectors. As the demand for EV infrastructure increases, there is a growing requirement for intelligent platforms that can efficiently handle charging operations, optimize energy usage, and provide smooth digital experiences for users.

AiElectron is a smart EV charging and energy management platform developed by Smart eMobility. The platform is designed for both residential and commercial users and offers solutions related to EV charging, smart energy monitoring, demand response participation, and utility management. It also supports advanced concepts such as predictive energy analytics, grid balancing, and Virtual Power Plant (VPP) integration.

The internship work mainly involved studying the overall working of the AiElectron platform and contributing towards improving platform workflows, usability, and feature efficiency from a techno-functional perspective. The work included understanding different modules, analysing user interactions, identifying workflow limitations, and suggesting improvements based on user and business requirements.

During the internship, exposure was gained to various areas such as Smart eMobility systems, EV charging operations, Energy Management Systems (EMS), charging station exploration modules, demand response events, and live energy monitoring. The internship also provided practical experience in activities like requirement analysis, workflow documentation, user story creation, feature enhancement planning, UI/UX evaluation, and coordination with development teams.

## **2. Problem Statement and Domain**

The growing adoption of EVs and increasing electricity demand have introduced several challenges in the utility and charging infrastructure industry.

Conventional systems often lack proper coordination between EV charging networks, energy optimization systems, and user-friendly digital platforms.

One of the key challenges is developing a unified solution that can efficiently support EV charging management, charging station discovery, energy tracking, and participation in smart utility programs while maintaining a simple and intuitive user experience. Many existing platforms also face issues such as complex workflows, scalability limitations, inefficient navigation structures, and poor visibility of energy-related data.

The Smart eMobility platform addresses these challenges by providing an integrated digital ecosystem that improves interaction between users, EV

infrastructure, and utility services. The platform supports functionalities such as:

- EV charging station discovery and operation
- Smart Energy Management System monitoring
- Demand Response participation and tracking
- Charging session and profile management
- Energy consumption analytics and visualization
- Real-time monitoring and reporting of energy events

The project primarily belongs to the Utility and Smart Energy domain, focusing on EV charging integration, intelligent energy management, demand optimization, and digital utility solutions.

The main objective during the internship was to enhance workflow efficiency, improve user experience, and contribute towards creating scalable platform solutions that can adapt to varying client and operational requirements. The work also involved understanding how smart utility platforms help in energy optimization, peak demand handling, and future smart grid technologies.

### **3. Techniques and Technologies Used**

The internship followed a techno-functional and product-focused approach aimed at analysing and improving digital utility platform operations. The work mainly involved workflow analysis, requirement understanding, UI/UX evaluation, feature enhancement planning, and functional coordination to improve platform usability and operational efficiency.

A major part of the internship involved studying existing workflows, identifying usability gaps, and proposing practical enhancements to improve both system performance and user interaction.

#### **Techniques Implemented**

##### **Workflow and Requirement Analysis**

Platform functionalities and workflows were studied from both business and operational perspectives to understand system behaviour, user interaction patterns, and functional requirements. This analysis helped in identifying improvement opportunities within the platform.

##### **User Experience and Navigation Study**

Different user journeys and navigation flows were reviewed to identify usability issues, unnecessary workflow complexity, and user pain points. The analysis focused on simplifying user interaction and improving accessibility.

##### **Feature Evaluation and Enhancement Planning**

Existing features were evaluated to identify future enhancement possibilities and workflow optimizations. Suggestions were prepared to improve scalability, platform efficiency, and overall user experience.

##### **User Story and Documentation Preparation**

Functional documentation, workflow descriptions, and user stories were created to clearly communicate requirements, expected behaviour, and enhancement

ideas among teams.

#### UI/UX Review and Optimization

The platform interface and workflows were analysed to improve readability, simplify navigation, reduce interface complexity, and create a more user-friendly experience.

#### Functional Validation and Team Coordination

Features and workflows were reviewed functionally to ensure they aligned with business goals and implementation expectations. Discussions and validations were conducted with relevant teams to maintain clarity and smooth execution.

#### Tools and Technologies Used

- Jira  
Used for task assignment, sprint management, issue tracking, and workflow monitoring.
- Confluence  
Used for maintaining documentation, requirement records, and project-related knowledge sharing.
- Figma  
Used for interface planning, workflow visualization, and UI/UX enhancement mockups.
- MS Office Tools  
Used for documentation, analysis reports, presentations, and project reporting.
- Product Management Practices  
Applied user-centric and techno-functional methodologies for workflow optimization, feature planning, stakeholder coordination, and scalable solution analysis.

## **4. Methodology**

The internship methodology followed a structured process for analysing and improving the digital utility platform. The workflow started with studying platform functionalities and understanding existing user workflows, followed by requirement analysis and identification of improvement areas.

Based on the findings, user stories, workflow documentation, UI/UX improvement suggestions, and feature enhancement plans were prepared to improve operational efficiency and user experience. The proposed improvements were then functionally validated and coordinated with the relevant teams to ensure proper alignment with implementation goals and business requirements.

